Youngji Koh

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youngji-koh.github.io

Summary

Researcher specializing in human–AI interaction, agentic and multi-agent systems, and multimodal computing for digital mental health and wellbeing.

Research Interest

Human–AI Interaction • Agentic and Multi-Agent Systems • Digital Mental Health • Multimodal Behavioral Modeling • LLM-based Contextual Reasoning • Ubiquitous Computing

Education

Ph.D. in School of Computing Sep. 2021 – Aug. 2026
KAIST, Daejeon, Republic of Korea
Interactive Computing Laboratory (Advisor: Dr. Uichin Lee)

Dissertation: Multimodal Behavior Modeling and AI Agent Systems for Mental Healthcare in Intelligent Environments.

M.S. in School of Computing Sep. 2019 – Aug. 2021
KAIST, Daejeon, Republic of Korea
Software Architecture Laboratory (Advisor: Dr. Sungwon Kang)

B.S. in Computer Science and Engineering Mar. 2015 – Aug. 2019
Ewha Womans University, Seoul, Republic of Korea
GPA: 4.02/4.30 (Summa Cum Laude)

Research Experience

Visiting Researcher (Postdoctoral Fellow) Sep. 2026 – Aug. 2027
Interactive Intelligent Systems Laboratory, The University of Tokyo (Host: Dr. Koji Yatani)

Fellowship: National Research Foundation of Korea (NRF) Postdoctoral Fellowship.

- Conducting research on human–AI interaction and AI-based mental healthcare systems.
- Investigating personalized and multi-agent AI approaches for supporting cognitive reframing, reflection, and mental wellbeing.

Undergraduate Research Intern May 2018 – May 2019
Human-Computer Interaction Laboratory, Ewha Womans University (Advisor: Dr. Uran Oh)

- Built a touchscreen-based mobile application providing object-level verbal descriptions to support independent artwork exploration by people with visual impairments.
- Designed a mobile application for supporting conflict management in romantic relationships on mobile messengers.

Funded Research Projects

Development of an Early Prediction System for Complex Diseases Jun. 2025 – Aug. 2026
Institute for ICT Planning & Evaluation (IITP), Lead Researcher

- Designed a comorbidity-aware framework that coordinates disorder-specific agents for diagnosis and intervention planning.
- Focused on integrating LLM-based reasoning, multimodal sensing, and digital health applications for early intervention.
- **Keywords:** LLM Agent · Human-Centered Computing · Multimodal Sensing · Digital Health · Mental Health & Wellbeing

AI Models for Multimodal Mental Health Prediction Sep. 2023 – Aug. 2025
LG Electronics, Lead Researcher

- Developed deep learning and machine learning models using IoT, mobile, and wearable data for depression and anxiety prediction.
- Evaluated detection accuracy and reliability through large-scale, real-world deployments.
- **Keywords:** Machine Learning · Mental Health & Wellbeing · Multimodal Sensing · Digital Health

IoT, Mobile, and Wearable Data Visualization for Mental Health Self-Reflection May 2023 – Feb. 2025
LG Electronics, Lead Researcher

- Collected multimodal behavioral and physiological data via IoT, mobile, and wearable devices.
- Designed and evaluated personal informatics dashboards to promote self-reflection and wellbeing.
- The study was featured in national media outlets, including Yonhap News, Chosun Ilbo, Herald Economy, and Digital Times.
- **Keywords:** Data Visualization · IoT & Wearables · Mental Health · Ubiquitous Computing

Robotic Counseling System for Mental Health Support Feb. 2023 – Oct. 2024
LG Electronics, Research Contributor

- Supported a two-week field study with 20 participants evaluating emotional awareness, engagement, and sense of place in robotic counseling.
- **Keywords:** Social Robots · Mental Health · Human-Robot Interaction · Digital Interventions

Context-Aware Multimodal Smart Speaker for Mental Health Self-Tracking Sep. 2021 – Sep. 2023
LG Electronics, Lead Researcher

- Designed and developed a context-aware multimodal smart speaker for in-home mental health self-tracking.
- Led in-the-wild deployment for continuous behavioral monitoring and physiological data collection.
- The project received national media coverage highlighting its contribution to digital mental health innovation, including Yonhap News, Herald Economy, Kookmin Ilbo, and Maeil Business.
- **Keywords:** Smart Speaker · Context-Aware Sensing · Smart Home · Digital Health

Publications

Preprints

[J2] **A Multimodal Sensor Fusion Approach for Mental Health Detection Using Mobile, Wearable, and IoT Sensors**
Youngji Koh, *Gyuna Kim, Chanhee Lee, Panyu Zhang, Yunhee Ku, Inhwan Choi, Jewoo Ryu, and Uichin Lee*
Under review at *IEEE Journal of Biomedical and Health Informatics (J-BHI)*. IEEE. *(Minor Revision Submitted)*

Journal & Conference

[C8] **Exploring Data-Driven Approaches to Stress Management: A Systematic Review of Stress Tracking, Intervention, and System Evaluation Methods**
Youngji Koh, *Gwangyoung Lee, Jeonghyun Kim, Yugyeong Jung, Hwajung Hong, and Uichin Lee*
In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)* (Barcelona, Spain, Apr. 13–17, 2026). CHI'26. ACM.

[C7] **Leveraging Smartphone Human Interaction Routine Behavior Task Mining and Modeling for Daily Stress Monitoring**
Hansoo Lee, Taehyeon Park, ***Youngji Koh****, Jaegil Lee, and Uichin Lee*
In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*. UbiComp'26. ACM.

[C6] **Harnessing Home IoT for Self-tracking Emotional Wellbeing: Behavioral Patterns, Self-Reflection, and Privacy Concerns**
Youngji Koh*, Chanhee Lee, Eunki Joung, Hyunsoo Lee, and Uichin Lee*
In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* (Espoo, Finland, Oct. 14–16, 2025). UbiComp'25. ACM.

[C5] **'In That Small Space with Just the Two of Us': User Experiences with Cumpa in a Robotic Counseling Center**
Chanhee Lee, Eunki Joung*,* ***Youngji Koh****, Esther Kim, Sohwi Son, Sunjung Kwon, and Uichin Lee (*: equal contribution)*
In *Proceedings of the ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)* (Bergen, Norway, Oct. 18–22, 2025). CSCW'25. ACM.

[C4] **Exploring Context-Aware Mental Health Self-Tracking Using Multimodal Smart Speakers in Home Environments**
Jieun Lim,* ***Youngji Koh****, Auk Kim, and Uichin Lee (*: equal contribution)*
In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)* (Honolulu, Hawaii, USA, May 11–16, 2024). CHI'24. ACM.

[C3] **Interrupting for Microlearning: Understanding Perceptions and Interruptibility of Proactive Conversational Microlearning Services**
Minyeong Kim, Jiwook Lee*,* ***Youngji Koh****, Chanhee Lee, Uichin Lee, and Auk Kim (*: equal contribution)*
In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)* (Honolulu, Hawaii, USA, May 11–16, 2024). CHI'24. ACM.

[J1] **Deep Learning-based Bug Report Summarization Using Sentence Significance Factors**
Youngji Koh*, Sungwon Kang, and Seonah Lee*
Applied Sciences, 2022.

[C2] **Bug Report Summarization using Believability Score and Text Ranking**
Youngji Koh*, Sungwon Kang, and Seonah Lee*
International Conference on Artificial Intelligence in Information and Communication (ICAIIC 2021). IEEE.

[C1] **Deep Learning Based Bug Report Summarization Using Sentence Assessment Score**
Youngji Koh *and Sungwon Kang*
Korea Software Congress (KSC 2020). *Best Paper Award*

Poster & Workshop

[W2] **Designing Clinician-Centered Explanations for Sensing-based Longitudinal Mental Health Monitoring**
Esther Kang, ***Youngji Koh****, Seungwan Jin, and Uichin Lee*
ACM UbiComp/ISWC 2026 XAI for U Workshop

[W1] **Data Visualization for Mental Health Monitoring in Smart Home Environment: A Case Study**
Youngji Koh*, Chanhee Lee, Yunhee Ku, and Uichin Lee*
IEEE VIS 2023 Workshop on Visual Analytics in Healthcare

[P4] **LV-Linker: Supporting Fine-grained User Interaction Analyses by Linking Smartphone Log and Recorded Video Data**
Hansoo Lee, Sangwook Lee, ***Youngji Koh****, and Uichin Lee*
ACM Symposium on User Interface Software and Technology (UIST 2022)

[P3] **Supporting Object-level Exploration of Artworks by Touch for People with Visual Impairments**
Nahyun Kwon, ***Youngji Koh****, and Uran Oh*
ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2019)

[P2] **Analysis of Conflicts in Romantic Relationships on Mobile Messengers**
Young-Ji Koh*, Hyunggu Jung, and Uran Oh*
HCI Korea 2019

[P1] **Real-time Fire Evacuation System based on Indoor Location Tracking and Route Optimization**
Young-Ji Koh, *Naeun Go, ChaeYoon Kim, SooHee Lee, and Sangsoo Park*
Korea Software Congress (KSC 2018). *Student Paper Competition, Merit Award*

Honors and Awards

Public Relations Award Dec. 2025
Donga Science, Korea

Creative Research Award Sep. 2025
K-Data Science Conference, Korea

Special Recognition for Outstanding Reviews Dec. 2024
ACM CHI

Ewha Womans University Scholarship Spring 2018
Merit-based scholarship

Teaching Experience

Teaching Assistant, KAIST Fall 2021, Spring 2023
CS492 Data Visualization

Teaching Assistant, KAIST Spring 2020
CS204 Discrete Mathematics

Community Service

HCI@KAIST 2024, 2025
Student Steering Committee

Academic Service

Reviewer
ACM Transactions on Computing for Healthcare
IEEE Transactions on Computational Social Systems
ACM UbiComp/ISWC Notes and Briefs
ACM CHI